AQA 3.1.4 and 3.1.10 Equilibrium constants K_c and K_p

AQA A-level Chemistry · Revision questions · Sheet 6 of 9

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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20 marks · about 25 minutes

Key information

- Units: substitute the units into the expression and cancel (e.g. kPa^{-2}, $mol^{-1}\ dm^3$ or no units)
- For the reverse reaction K' = 1/K; halving the equation gives √K
- Mole fraction = moles of that gas ÷ total moles of gas
- Partial pressure = mole fraction × total pressure
- K_p uses partial pressures in round brackets, e.g. $K_p = p(C)^2 \div p(A)\ p(B)^3$; never square brackets

Section A: Short-answer questions

01 For an endothermic equilibrium, state the effect of decreasing the temperature on the value of K_c and on the equilibrium yield of products.

[2 marks]

02 $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$

2.0 mol of carbon monoxide is mixed with 3.0 mol of hydrogen and allowed to reach equilibrium. The equilibrium mixture contains 0.6 mol of methanol. What is the total amount, in mol, of gas at equilibrium?

[1 mark]

A 3.2
B 3.8
C 4.4
D 5.0

03 Deduce the units of K_p for the equilibrium $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ when partial pressures are measured in kPa.

[1 mark]

04 Which statement about the addition of a catalyst to an equilibrium mixture is correct?

[1 mark]

A The activation energy for the reverse reaction increases.
B The equilibrium constant for the forward reaction increases.
C The rate of the reverse reaction increases.
D The enthalpy change for the forward reaction decreases.

Section B: Longer questions

05 Ethanol can be made from ethene and steam.

$C_2H_4(g) + H_2O(g) \rightleftharpoons CH_3CH_2OH(g) \quad \Delta H = -45\ kJ\ mol^{-1}$

The table shows the mole fractions of each of the gases in an equilibrium mixture at 6000 kPa.

Gas	Mole fraction
Ethene	0.645

Gas	Mole fraction
Steam	0.323
Ethanol	0.0321

05.1 Give an expression for K_p for this reaction. Calculate the value of K_p at 6000 kPa. State the units.

[4 marks]

05.2 State the effect, if any, of an increase in volume of the container on the value of K_p for this reaction at a constant temperature.

[1 mark]

06 Sulfur trioxide decomposes on heating to form an equilibrium mixture containing sulfur dioxide and oxygen.

$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$

06.1 A sample of sulfur trioxide was heated and allowed to reach equilibrium at a given temperature. The equilibrium mixture contained 6.08 g of sulfur dioxide. Calculate the mass, in g, of oxygen gas in the equilibrium mixture.

[2 marks]

06.2 A different mass of sulfur trioxide was heated and allowed to reach equilibrium at 1050 K. The amounts of each substance in the equilibrium mixture are shown in the table.

Substance	Amount at equilibrium / mol
sulfur trioxide	0.320
sulfur dioxide	1.20
oxygen	0.600

For this reaction at 1050 K, $K_p = 7.62 \times 10^5$ Pa

Calculate the mole fraction of each substance at equilibrium. Give the expression for K_p. Calculate the total pressure, in Pa, of this equilibrium mixture.

[4 marks]

06.3 At 1050 K, $K_p = 7.62 \times 10^5$ Pa. At 500 K, $K_p = 3.94 \times 10^4$ Pa. Explain how this information can be used to deduce that the forward reaction is endothermic.

[2 marks]

06.4 Use the value of K_p at 500 K to calculate the value of K_p, at 500 K, for the equilibrium $SO_3(g) \rightleftharpoons SO_2(g) + \frac{1}{2}O_2(g)$. Deduce the units of K_p.

[2 marks]

Mark Scheme

M1, M2... are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	K_c decreases yield of products decreases	*Source: Practice question (written for this set)*	2
02	**B**	*Source: 7404/2 Jun 2022 Q19*	1
03	kPa	*Source: Practice question (written for this set)*	1
04	**C**	*Source: 7405/3 Nov 2021 Q20*	1
05.1	M1 $K_p = p(CH_3CH_2OH) \div (p(C_2H_4) \times p(H_2O))$ M2 = (0.0321 × 6000) ÷ ((0.645 × 6000) × (0.323 × 6000)) = 192.6 ÷ (3870 × 1938) M3 $K_p = 2.56 \times 10^{-5}$ M4 kPa^{-1}	*Source: 7405/1 Jun 2024 Q04.4–04.5* Penalise square brackets M3 allow 2.57×10^{-5}	4
05.2	No effect		1
06.1	M1 $n(SO_2) = 6.08 \div 64.1 = 0.0949$ so $n(O_2) = 0.0474$ M2 mass $O_2 = 0.0474 \times 32.0 = 1.52$ g	*Source: 7405/1 Jun 2019 Q07*	2
06.2	M1 mole fractions $SO_3 = 0.15$, $SO_2 = 0.57$, $O_2 = 0.28$ M2 $K_p = p(SO_2)^2 \times p(O_2) \div p(SO_3)^2$ M3 $P = K_p \times 0.15^2 \div (0.57^2 \times 0.28)$ M4 $P = 1.91 \times 10^5$ Pa	Do not accept [ ] Allow 1.88×10^5 to 1.94×10^5	4
06.3	M1 K_p is higher at higher temperature M2 at higher temperature more dissociation occurs / more products / equilibrium shifts to the right	M2 dependent on M1	2
06.4	M1 $\sqrt{(3.94 \times 10^4)} = 198.5$ M2 $Pa^{½}$	If $\sqrt{(7.62 \times 10^5)} = 873$ lose M1 but allow M2	2